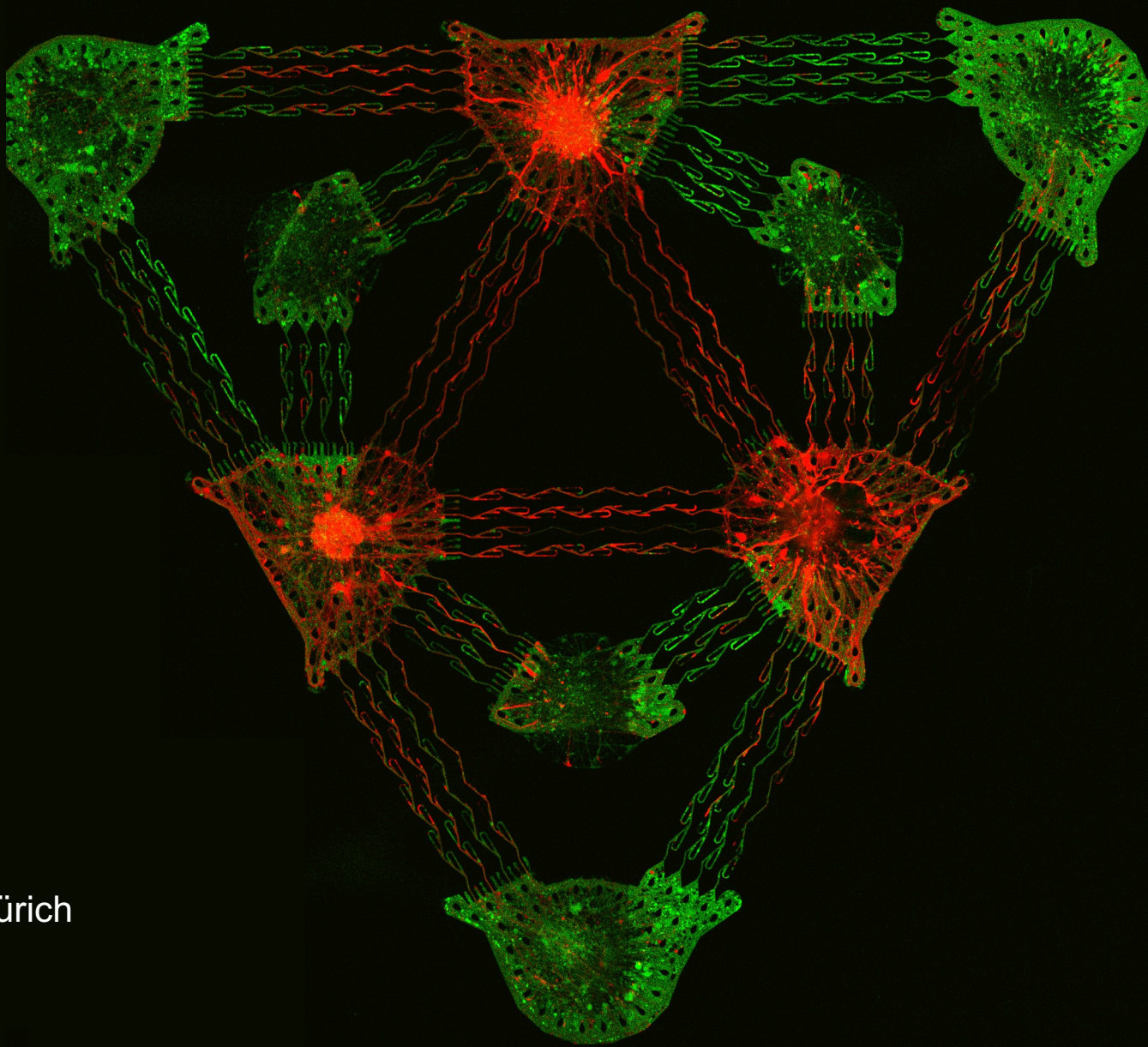




Introduction to the final task

P&S 2026

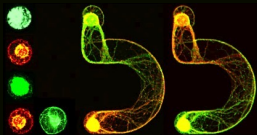
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Final Dataset

Final week

- Base frequency $F_{base} = 0.5\text{Hz}$
- Sweep 1-40Hz, step-size: 1Hz, repeated 60 times
- Patterns are applied in a random order
- After each stimulation train, wait 1s
- Per-task, data comes from one network
- All groups receive the same training and test data.

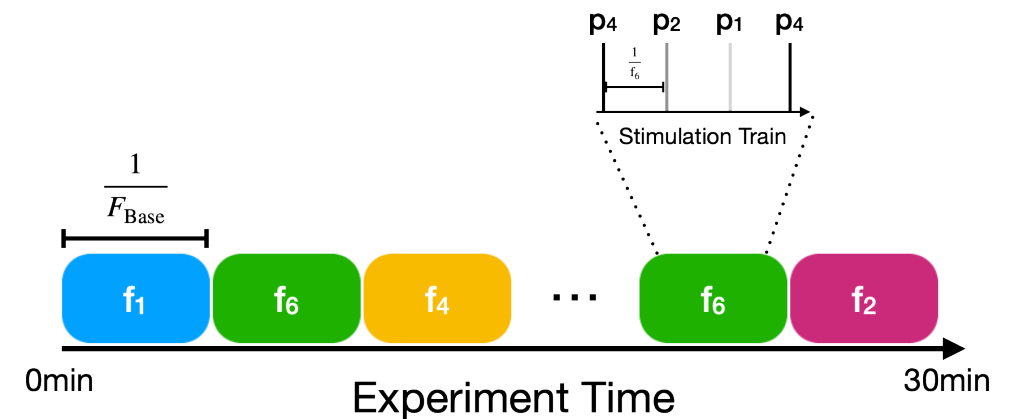
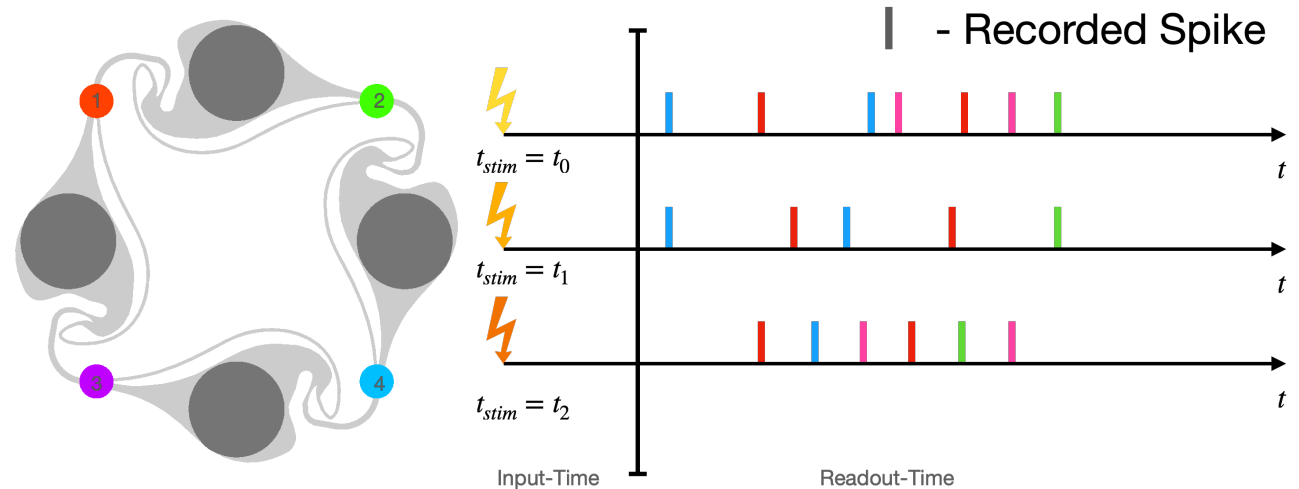
4 Files are available:

Task 1:

- **Training data** – Both stimulation parameters and responses are provided.
- **Test data** – Only stimulation responses are provided.

Task 2:

- **Training data** – Both stimulation parameters and responses are provided.
- **Test data** – Only stimulation parameters are provided.



How to access the data? How to send your responses?

Example of how to access the data is on the Master branch of `pands_ibnnwml` gitlab.

```
Network = 5 # or 6 for pattern prediction
DIV      = 40
group_data = False
data_train = load_data(Network, DIV, group_data)
data_test = load_data(Network, DIV, group_data, test_mode=True)
# All your processing goes here, ensure you do not mix up the order for your prediction
save_data(
    Network,
    DIV,
    test_mode=True,
    binned_spike_train_responses=None, # or your responses
    stimulation_patterns=your_prediction # or None if you predict the responses
)
```

Shape: Number of
samples x electrodes
x time bins (Task 2)

Number of samples
(Task 1)

Evaluation metrics

For **Task 1**, we will use F1 score as a primary metric: $F1 = \frac{2 \cdot TP}{2 \cdot TP + FP + FN}$. F1 = 1 means that all true positives were identified and no false positives were detected. F1 = 0 means the opposite (the worst outcome).

Task 2

For each stimulation pattern and each electrode, we will compute an *empirical PSTH*: the average spike count in 1 ms bins across the repetitions of that pattern and frequency. We will do this for both the true responses and the model's predicted responses.

Each PSTH is normalized to a probability distribution over time bins. For each (electrode, pattern+frequency), we compute the Wasserstein-1 distance (in ms) between the predicted and true normalized PSTHs. The final score is the weighted mean Wasserstein distance across all included (electrode, pattern+frequency) pairs.

Send us the predictions on both test data by Sunday midnight so that we can do the ranking in time.

Next week: presentations

We will have a presentation session next week, starting at 14:15 at GLC D28.

Each group has 10 minutes to present their work.

Content of the presentation:

1. **Explanation of the models:** What methods did you use? What are their key mechanisms? What assumptions did you make?
2. **Results:** What was the performance of the models? Why did they work/fail? Discuss performance on the test data using your preferred evaluation metrics.

After the presentations, we will discuss the group ranking using the model evaluation from the previous slide and award the winners.